Lesson 15: Neyman-Pearson Lemma; Likelihood Ratio Tests

BSTA 551: Statistical Inference

Jessica Minnier

2026-02-18

Lesson 15: Neyman-Pearson Lemma; Likelihood Ratio Tests

Review: Where We Left Off

Lesson 14: P-Values; Power and Sample Size Determination

- P-values are Uniform(0,1) under H_0 by the probability integral transformation
- Power depends on the noncentrality parameter $\delta = (\mu' - \mu_0)/(\sigma/\sqrt{n})$
- Sample size determination via `power.t.test()` for planning clinical trials
- Multiple testing adjustments: Bonferroni (FWER) and Benjamini-Hochberg (FDR)

Review: Where We Left Off

Lesson 14: P-Values; Power and Sample Size Determination

- P-values are Uniform(0,1) under H_0 by the probability integral transformation
- Power depends on the noncentrality parameter $\delta = (\mu' - \mu_0)/(\sigma/\sqrt{n})$
- Sample size determination via `power.t.test()` for planning clinical trials
- Multiple testing adjustments: Bonferroni (FWER) and Benjamini-Hochberg (FDR)

Today's Goals:

1. Understand the concept of a **most powerful** test for simple hypotheses
2. State and prove the **Neyman-Pearson Lemma**
3. Define **power functions** and **uniformly most powerful (UMP)** tests
4. Develop the **likelihood ratio test (LRT)** for composite hypotheses
5. Connect the LRT to familiar tests (z test, t test)
6. Apply the large-sample chi-squared approximation for LRTs

Roadmap for Today

Topic	Key Idea	Textbook
Simple hypotheses	Both H_0 and H_a fully specify the distribution	Devore 9.5
Neyman-Pearson Lemma	The likelihood ratio test is most powerful for simple H_0 vs simple H_a	Devore 9.5
Power function	$\pi(\theta') = P_{\theta'}(\text{Reject } H_0)$ as a function of θ	Devore 9.5
UMP tests	Most powerful for every value in the alternative	Devore 9.5
Likelihood ratio tests	General principle for composite hypotheses	Devore 9.5
LRT $\rightarrow$ t test	The one-sample t test is a LRT	Devore 9.5
Large-sample LRT	$-2\ln(\Lambda) \xrightarrow{d} \chi^2_m$	Devore 9.5

Roadmap for Today

Topic	Key Idea	Textbook
Simple hypotheses	Both H_0 and H_a fully specify the distribution	Devore 9.5
Neyman-Pearson Lemma	The likelihood ratio test is most powerful for simple H_0 vs simple H_a	Devore 9.5
Power function	$\pi(\theta') = P_{\theta'}(\text{Reject } H_0)$ as a function of θ	Devore 9.5
UMP tests	Most powerful for every value in the alternative	Devore 9.5
Likelihood ratio tests	General principle for composite hypotheses	Devore 9.5
LRT $\rightarrow$ t test	The one-sample t test is a LRT	Devore 9.5
Large-sample LRT	$-2\ln(\Lambda) \xrightarrow{d} \chi^2_m$	Devore 9.5

Why This Matters

The Neyman-Pearson Lemma provides the **theoretical foundation** for optimal hypothesis testing. It tells us *why* the tests we have been using (z tests, t tests) are the best possible tests in their respective settings. The likelihood ratio principle then gives us a **general recipe** for constructing tests in new situations.

Simple Hypotheses and Optimal Tests (Devore 9.5)

Simple vs. Composite Hypotheses

Recall from Lesson 12:

Definition (Simple and Composite Hypotheses)

A hypothesis is **simple** if it completely specifies the distribution of the sample $X_1, \ldots, X_n$. Otherwise, it is **composite**.

Simple vs. Composite Hypotheses

Recall from Lesson 12:

> **Definition (Simple and Composite Hypotheses)**
>
> A hypothesis is **simple** if it completely specifies the distribution of the sample $X_1, \ldots, X_n$. Otherwise, it is **composite**.

Examples:

- $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Exp}(\lambda)$:
  - $H : \lambda = 5$ is **simple** (distribution fully determined)
  - $H : \lambda < 5$ is **composite** (many possible distributions)

Simple vs. Composite Hypotheses

Recall from Lesson 12:

Definition (Simple and Composite Hypotheses)

A hypothesis is **simple** if it completely specifies the distribution of the sample $X_1, \ldots, X_n$. Otherwise, it is **composite**.

Examples:

- $X_1, \ldots, X_n \overset{iid}{\sim} \text{Exp}(\lambda)$:
  - $H : \lambda = 5$ is **simple** (distribution fully determined)
  - $H : \lambda < 5$ is **composite** (many possible distributions)
- $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ **known**:
  - $H : \mu = 100$ is **simple**
  - $H : \mu \neq 100$ is **composite**

Simple vs. Composite Hypotheses

Recall from Lesson 12:

Definition (Simple and Composite Hypotheses)

A hypothesis is **simple** if it completely specifies the distribution of the sample $X_1, \ldots, X_n$. Otherwise, it is **composite**.

Examples:

- $X_1, \ldots, X_n \overset{iid}{\sim} \text{Exp}(\lambda)$:
  - $H : \lambda = 5$ is **simple** (distribution fully determined)
  - $H : \lambda < 5$ is **composite** (many possible distributions)
- $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ **known**:
  - $H : \mu = 100$ is **simple**
  - $H : \mu \neq 100$ is **composite**
- $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ **unknown**:
  - $H : \mu = 100$ is **composite** (because σ is unspecified — it is a *nuisance parameter*)

The Optimality Question

We have developed many test procedures: z tests, t tests, score tests, exact binomial tests...

The Optimality Question

We have developed many test procedures: z tests, t tests, score tests, exact binomial tests...

But are these the *best* tests?

The Optimality Question

We have developed many test procedures: z tests, t tests, score tests, exact binomial tests...

But are these the *best* tests?

For a test of $H_0 : \theta = \theta_0$ vs $H_a : \theta = \theta_a$, both simple, there are infinitely many possible rejection regions R. Each one gives:

- A **Type I error rate**: $\alpha = P_{\theta_0}((X_1, \ldots, X_n) \in R)$
- A **Type II error rate**: $\beta = P_{\theta_a}((X_1, \ldots, X_n) \notin R)$

The Optimality Question

We have developed many test procedures: z tests, t tests, score tests, exact binomial tests…

But are these the *best* tests?

For a test of $H_0 : \theta = \theta_0$ vs $H_a : \theta = \theta_a$, both simple, there are infinitely many possible rejection regions R. Each one gives:

- A **Type I error rate**: $\alpha = P_{\theta_0}((X_1, \ldots, X_n) \in R)$
- A **Type II error rate**: $\beta = P_{\theta_a}((X_1, \ldots, X_n) \notin R)$

Goal: Among all tests with $\alpha \leq \alpha^*$ (controlling Type I error), find the test that **minimizes** β (equivalently, **maximizes power**).

The Optimality Question

We have developed many test procedures: z tests, t tests, score tests, exact binomial tests...

But are these the *best* tests?

For a test of $H_0 : \theta = \theta_0$ vs $H_a : \theta = \theta_a$, both simple, there are infinitely many possible rejection regions R. Each one gives:

- A **Type I error rate**: $\alpha = P_{\theta_0}((X_1, \ldots, X_n) \in R)$
- A **Type II error rate**: $\beta = P_{\theta_a}((X_1, \ldots, X_n) \notin R)$

Goal: Among all tests with $\alpha \leq \alpha^*$ (controlling Type I error), find the test that **minimizes** β (equivalently, **maximizes power**).

Medical Research Motivation

In a clinical trial comparing a new drug to a standard, we want the test that gives us the **best chance of detecting a real treatment effect** (high power) while maintaining the required significance level. The Neyman-Pearson Lemma tells us exactly how to construct such a test.

The Likelihood Ratio as a Test Statistic

Intuition: Given data $x_1, \ldots, x_n$, which hypothesis makes the data more likely?

The Likelihood Ratio as a Test Statistic

Intuition: Given data $x_1, \ldots, x_n$, which hypothesis makes the data more likely?

Consider the ratio:

$$\Lambda^* = \frac{f(x_1, \ldots, x_n; \theta_a)}{f(x_1, \ldots, x_n; \theta_0)} = \frac{L(\theta_a)}{L(\theta_0)}$$

The Likelihood Ratio as a Test Statistic

Intuition: Given data $x_1, \ldots, x_n$, which hypothesis makes the data more likely?

Consider the ratio:

$$\Lambda^* = \frac{f(x_1, \ldots, x_n; \theta_a)}{f(x_1, \ldots, x_n; \theta_0)} = \frac{L(\theta_a)}{L(\theta_0)}$$

- If $\Lambda^* \gg 1$: data much more consistent with $H_a \Rightarrow$ evidence against H_0
- If $\Lambda^* \approx 1$: data equally consistent with both $\Rightarrow$ no strong evidence
- If $\Lambda^* \ll 1$: data more consistent with $H_0 \Rightarrow$ no evidence against H_0

The Likelihood Ratio as a Test Statistic

Intuition: Given data $x_1, \ldots, x_n$, which hypothesis makes the data more likely?

Consider the ratio:

$$\Lambda^* = \frac{f(x_1, \ldots, x_n; \theta_a)}{f(x_1, \ldots, x_n; \theta_0)} = \frac{L(\theta_a)}{L(\theta_0)}$$

- If $\Lambda^* \gg 1$: data much more consistent with $H_a \Rightarrow$ evidence against H_0
- If $\Lambda^* \approx 1$: data equally consistent with both $\Rightarrow$ no strong evidence
- If $\Lambda^* \ll 1$: data more consistent with $H_0 \Rightarrow$ no evidence against H_0

Natural test: Reject H_0 when $\Lambda^* \geq k$ for some constant $k > 0$.

The Likelihood Ratio as a Test Statistic

Intuition: Given data $x_1, \ldots, x_n$, which hypothesis makes the data more likely?

Consider the ratio:

$$\Lambda^* = \frac{f(x_1, \ldots, x_n; \theta_a)}{f(x_1, \ldots, x_n; \theta_0)} = \frac{L(\theta_a)}{L(\theta_0)}$$

- If $\Lambda^* \gg 1$: data much more consistent with $H_a \Rightarrow$ evidence against H_0
- If $\Lambda^* \approx 1$: data equally consistent with both $\Rightarrow$ no strong evidence
- If $\Lambda^* \ll 1$: data more consistent with $H_0 \Rightarrow$ no evidence against H_0

Natural test: Reject H_0 when $\Lambda^* \geq k$ for some constant $k > 0$.

Convention Note

Devore defines the NP ratio as $L(\theta_a)/L(\theta_0)$ (reject when large). Chihara defines it as $L(\theta_0)/L(\theta_a)$ (reject when small). Both formulations are equivalent — just inverted. We will primarily follow Devore's convention in the statement of the lemma.

The Neyman-Pearson Lemma

Statement of the Neyman-Pearson Lemma

The Neyman-Pearson Lemma (Devore 9.5)

For testing a simple null hypothesis $H_0 : \theta = \theta_0$ versus a simple alternative $H_a : \theta = \theta_a$, let k be a fixed positive number and form the rejection region

$$R^* = \left\{ (x_1, \dots, x_n) : \frac{f(x_1, \dots, x_n; \theta_a)}{f(x_1, \dots, x_n; \theta_0)} \geq k \right\}$$

Let $\alpha^* = P((X_1, \dots, X_n) \in R^* \text{ when } \theta = \theta_0)$ and let β^* denote the Type II error probability.

Then for **any other** test procedure with Type I error probability $\alpha \leq \alpha^*$, the Type II error probability must satisfy $\beta \geq \beta^*$.

Statement of the Neyman-Pearson Lemma

The Neyman-Pearson Lemma (Devore 9.5)

For testing a simple null hypothesis $H_0 : \theta = \theta_0$ versus a simple alternative $H_a : \theta = \theta_a$, let k be a fixed positive number and form the rejection region

$$R^* = \left\{(x_1, \ldots, x_n) : \frac{f(x_1, \ldots, x_n; \theta_a)}{f(x_1, \ldots, x_n; \theta_0)} \geq k\right\}$$

Let $\alpha^* = P((X_1, \ldots, X_n) \in R^* \text{ when } \theta = \theta_0)$ and let β^* denote the Type II error probability.

Then for **any other** test procedure with Type I error probability $\alpha \leq \alpha^*$, the Type II error probability must satisfy $\beta \geq \beta^*$.

In plain language: The test based on the likelihood ratio

- Has the **smallest** β (Type II error) among all tests with $\alpha \leq \alpha^*$
- Equivalently, has the **largest power** at θ_a among all level-α^* tests
- Is called the **most powerful (MP)** test at level α^*

Proof of the Neyman-Pearson Lemma

Let R be the rejection region of **any** test, so that:

$$\alpha = \sum_R f(x_1, \ldots, x_n; \theta_0), \quad \beta = 1 - \sum_R f(x_1, \ldots, x_n; \theta_a)$$

Proof of the Neyman-Pearson Lemma

Let R be the rejection region of **any** test, so that:

$$\alpha = \sum_R f(x_1, \ldots, x_n; \theta_0), \quad \beta = 1 - \sum_R f(x_1, \ldots, x_n; \theta_a)$$

For any constant $k > 0$, consider the linear combination $k\alpha + \beta$:

$$k\alpha + \beta = k \sum_R f(\mathbf{x}; \theta_0) + 1 - \sum_R f(\mathbf{x}; \theta_a) = 1 + \sum_R [k \cdot f(\mathbf{x}; \theta_0) - f(\mathbf{x}; \theta_a)]$$

Proof of the Neyman-Pearson Lemma

Let R be the rejection region of **any** test, so that:

$$\alpha = \sum_R f(x_1, \ldots, x_n; \theta_0), \quad \beta = 1 - \sum_R f(x_1, \ldots, x_n; \theta_a)$$

For any constant $k > 0$, consider the linear combination $k\alpha + \beta$:

$$k\alpha + \beta = k \sum_R f(\mathbf{x}; \theta_0) + 1 - \sum_R f(\mathbf{x}; \theta_a) = 1 + \sum_R [k \cdot f(\mathbf{x}; \theta_0) - f(\mathbf{x}; \theta_a)]$$

Key insight: The expression $k \cdot f(\mathbf{x}; \theta_0) - f(\mathbf{x}; \theta_a)$ can be positive or negative. To **minimize** $k\alpha + \beta$, choose R to include exactly the points where this expression is negative:

$$R^* = \{\mathbf{x} : k \cdot f(\mathbf{x}; \theta_0) - f(\mathbf{x}; \theta_a) \leq 0\} = \left\{\mathbf{x} : \frac{f(\mathbf{x}; \theta_a)}{f(\mathbf{x}; \theta_0)} \geq k\right\}$$

Proof of the Neyman-Pearson Lemma

Let R be the rejection region of **any** test, so that:

$$\alpha = \sum_R f(x_1, \ldots, x_n; \theta_0), \quad \beta = 1 - \sum_R f(x_1, \ldots, x_n; \theta_a)$$

For any constant $k > 0$, consider the linear combination $k\alpha + \beta$:

$$k\alpha + \beta = k \sum_R f(\mathbf{x}; \theta_0) + 1 - \sum_R f(\mathbf{x}; \theta_a) = 1 + \sum_R [k \cdot f(\mathbf{x}; \theta_0) - f(\mathbf{x}; \theta_a)]$$

Key insight: The expression $k \cdot f(\mathbf{x}; \theta_0) - f(\mathbf{x}; \theta_a)$ can be positive or negative. To **minimize** $k\alpha + \beta$, choose R to include exactly the points where this expression is negative:

$$R^* = \{\mathbf{x} : k \cdot f(\mathbf{x}; \theta_0) - f(\mathbf{x}; \theta_a) \leq 0\} = \left\{\mathbf{x} : \frac{f(\mathbf{x}; \theta_a)}{f(\mathbf{x}; \theta_0)} \geq k\right\}$$

This is exactly the Neyman-Pearson rejection region! So R^* minimizes $k\alpha + \beta$.

Proof (continued)

We've shown: $k\alpha^* + \beta^* \leq k\alpha + \beta$ for **all** rejection regions R.

Proof (continued)

We've shown: $k\alpha^* + \beta^* \leq k\alpha + \beta$ for **all** rejection regions R.

Now, for any test with $\alpha \leq \alpha^*$:

$$k\alpha^* + \beta^* \leq k\alpha + \beta$$

$$\beta^* \leq \beta + k(\alpha - \alpha^*) \leq \beta + k \cdot 0 = \beta$$

Proof (continued)

We've shown: $k\alpha^* + \beta^* \leq k\alpha + \beta$ for **all** rejection regions R.

Now, for any test with $\alpha \leq \alpha^*$:

$$k\alpha^* + \beta^* \leq k\alpha + \beta$$

$$\beta^* \leq \beta + k(\alpha - \alpha^*) \leq \beta + k \cdot 0 = \beta$$

since $\alpha - \alpha^* \leq 0$ and $k > 0$.

Proof (continued)

We've shown: $k\alpha^* + \beta^* \leq k\alpha + \beta$ for **all** rejection regions R.

Now, for any test with $\alpha \leq \alpha^*$:

$$k\alpha^* + \beta^* \leq k\alpha + \beta$$

$$\beta^* \leq \beta + k(\alpha - \alpha^*) \leq \beta + k \cdot 0 = \beta$$

since $\alpha - \alpha^* \leq 0$ and $k > 0$.

Therefore $\beta^* \leq \beta$ for all tests with $\alpha \leq \alpha^*$. ■

Proof (continued)

We've shown: $k\alpha^* + \beta^* \leq k\alpha + \beta$ for **all** rejection regions R.

Now, for any test with $\alpha \leq \alpha^*$:

$$k\alpha^* + \beta^* \leq k\alpha + \beta$$

$$\beta^* \leq \beta + k(\alpha - \alpha^*) \leq \beta + k \cdot 0 = \beta$$

since $\alpha - \alpha^* \leq 0$ and $k > 0$.

Therefore $\beta^* \leq \beta$ for all tests with $\alpha \leq \alpha^*$. ■

What Does This Mean?

The proof works by showing that the NP rejection region simultaneously minimizes **both** α and β in a trade-off sense. You cannot do better on one without doing worse on the other. The NP test is on the **efficient frontier** of the (α, β) trade-off.

Example: Adverse Events in a Drug Trial (Normal, Known σ)

A pharmaceutical company monitors a biomarker level in patients. Under the current treatment, biomarker levels follow $N(\mu, 1)$. An elevated biomarker signals a potential adverse event.

Test $H_0 : \mu = 5$ vs $H_a : \mu = 6$ based on n patients (known $\sigma = 1$).

Example: Adverse Events in a Drug Trial (Normal, Known σ)

A pharmaceutical company monitors a biomarker level in patients. Under the current treatment, biomarker levels follow $N(\mu, 1)$. An elevated biomarker signals a potential adverse event.

Test $H_0 : \mu = 5$ vs $H_a : \mu = 6$ based on n patients (known $\sigma = 1$).

Applying the Neyman-Pearson Lemma:

$$\frac{L(\mu_a = 6)}{L(\mu_0 = 5)} = \frac{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-(x_i-6)^2/2}}{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-(x_i-5)^2/2}} = e^{\sum(x_i-6)^2/2\cdot(-1)+\sum(x_i-5)^2/2}$$

Example: Adverse Events in a Drug Trial (Normal, Known σ)

A pharmaceutical company monitors a biomarker level in patients. Under the current treatment, biomarker levels follow $N(\mu, 1)$. An elevated biomarker signals a potential adverse event.

Test $H_0 : \mu = 5$ vs $H_a : \mu = 6$ based on n patients (known $\sigma = 1$).

Applying the Neyman-Pearson Lemma:

$$\frac{L(\mu_a = 6)}{L(\mu_0 = 5)} = \frac{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-(x_i-6)^2/2}}{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-(x_i-5)^2/2}} = e^{\sum(x_i-6)^2/2 \cdot (-1) + \sum(x_i-5)^2/2}$$

Simplifying the exponent:

$$= e^{(6-5)\sum x_i - n(6^2-5^2)/2} = e^{\sum x_i - 11n/2}$$

Example: Normal (continued)

The rejection region $\Lambda^* \geq k$ becomes:

$$e^{\sum x_i - 11n/2} \geq k$$

Example: Normal (continued)

The rejection region $\Lambda^* \geq k$ becomes:

$$e^{\sum x_i - 11n/2} \geq k$$

Taking logarithms (since $\mu_a - \mu_0 = 1 > 0$, the exponential is increasing in $\sum x_i$):

$$\sum x_i \geq k' \quad \iff \quad \bar{x} \geq k'' \quad \iff \quad \frac{\bar{x} - 5}{1/\sqrt{n}} \geq c$$

Example: Normal (continued)

The rejection region $\Lambda^* \geq k$ becomes:

$$e^{\sum x_i - 11n/2} \geq k$$

Taking logarithms (since $\mu_a - \mu_0 = 1 > 0$, the exponential is increasing in $\sum x_i$):

$$\sum x_i \geq k' \quad \iff \quad \bar{x} \geq k'' \quad \iff \quad \frac{\bar{x} - 5}{1/\sqrt{n}} \geq c$$

Setting $c = z_\alpha$, this is exactly the **one-sample z test**!

Example: Normal (continued)

The rejection region $\Lambda^* \geq k$ becomes:

$$e^{\sum x_i - 11n/2} \geq k$$

Taking logarithms (since $\mu_a - \mu_0 = 1 > 0$, the exponential is increasing in $\sum x_i$):

$$\sum x_i \geq k' \quad \Longleftrightarrow \quad \bar{x} \geq k'' \quad \Longleftrightarrow \quad \frac{\bar{x} - 5}{1/\sqrt{n}} \geq c$$

Setting $c = z_\alpha$, this is exactly the **one-sample z test**!

Key Result (Devore Example 9.24)

For $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ known, testing $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a$ where $\mu_a > \mu_0$:

The Neyman-Pearson Lemma shows that the **one-sample z test** (reject when $z \geq z_\alpha$) is the **most powerful** test at level α. Our familiar z test is optimal!

Example: Poisson Adverse Event Counts (Devore Example 9.23)

In post-market drug surveillance, the number of adverse events per reporting period is monitored. Suppose $X_1, \ldots, X_5 \overset{iid}{\sim} \text{Poisson}(\lambda)$ represent adverse event counts over 5 periods.

Test $H_0 : \lambda = 1$ (baseline rate) vs $H_a : \lambda = 2$ (elevated rate).

Example: Poisson Adverse Event Counts (Devore Example 9.23)

In post-market drug surveillance, the number of adverse events per reporting period is monitored. Suppose $X_1, \ldots, X_5 \overset{iid}{\sim} \text{Poisson}(\lambda)$ represent adverse event counts over 5 periods.

Test $H_0 : \lambda = 1$ (baseline rate) vs $H_a : \lambda = 2$ (elevated rate).

The Poisson likelihood is: $f(x_1, \ldots, x_5; \lambda) = e^{-5\lambda}\lambda^{\sum x_i} / \prod x_i!$

Example: Poisson Adverse Event Counts (Devore Example 9.23)

In post-market drug surveillance, the number of adverse events per reporting period is monitored. Suppose $X_1, \dots, X_5 \overset{iid}{\sim} \text{Poisson}(\lambda)$ represent adverse event counts over 5 periods.

Test $H_0 : \lambda = 1$ (baseline rate) vs $H_a : \lambda = 2$ (elevated rate).

The Poisson likelihood is: $f(x_1, \dots, x_5; \lambda) = e^{-5\lambda} \lambda^{\sum x_i} / \prod x_i!$

The likelihood ratio:

$$\frac{L(\lambda = 2)}{L(\lambda = 1)} = \frac{e^{-10} \cdot 2^{\sum x_i}}{e^{-5} \cdot 1^{\sum x_i}} = e^{-5} \cdot 2^{\sum x_i} \geq k$$

Example: Poisson Adverse Event Counts (Devore Example 9.23)

In post-market drug surveillance, the number of adverse events per reporting period is monitored. Suppose $X_1, \ldots, X_5 \overset{iid}{\sim} \text{Poisson}(\lambda)$ represent adverse event counts over 5 periods.

Test $H_0 : \lambda = 1$ (baseline rate) vs $H_a : \lambda = 2$ (elevated rate).

The Poisson likelihood is: $f(x_1, \ldots, x_5; \lambda) = e^{-5\lambda}\lambda^{\sum x_i} / \prod x_i!$

The likelihood ratio:

$$\frac{L(\lambda = 2)}{L(\lambda = 1)} = \frac{e^{-10} \cdot 2^{\sum x_i}}{e^{-5} \cdot 1^{\sum x_i}} = e^{-5} \cdot 2^{\sum x_i} \geq k$$

Taking logarithms: $\sum x_i \geq c$ where $c = \ln(k \cdot e^5) / \ln(2)$.

Example: Poisson (continued)

Finding the critical value: Under $H_0, Y = \sum X_i \sim \text{Poisson}(5)$.

Example: Poisson (continued)

Finding the critical value: Under $H_0, Y = \sum X_i \sim \text{Poisson}(5)$.

If we use $c = 10$: $\alpha^* = P(Y \geq 10 \mid Y \sim \text{Poisson}(5)) = 1 - 0.968 = 0.032$

If we use $c = 9$: $\alpha^* = P(Y \geq 9 \mid Y \sim \text{Poisson}(5)) = 1 - 0.932 = 0.068$

Example: Poisson (continued)

Finding the critical value: Under $H_0, Y = \sum X_i \sim \text{Poisson}(5)$.

If we use $c = 10$: $\alpha^* = P(Y \geq 10 \mid Y \sim \text{Poisson}(5)) = 1 - 0.968 = 0.032$

If we use $c = 9$: $\alpha^* = P(Y \geq 9 \mid Y \sim \text{Poisson}(5)) = 1 - 0.932 = 0.068$

For $\alpha \leq 0.05$, we use $c = 10$, giving $\alpha^* = 0.032$.

Example: Poisson (continued)

Finding the critical value: Under $H_0, Y = \sum X_i \sim \text{Poisson}(5)$.

If we use $c = 10$: $\alpha^* = P(Y \geq 10 \mid Y \sim \text{Poisson}(5)) = 1 - 0.968 = 0.032$

If we use $c = 9$: $\alpha^* = P(Y \geq 9 \mid Y \sim \text{Poisson}(5)) = 1 - 0.932 = 0.068$

For $\alpha \leq 0.05$, we use $c = 10$, giving $\alpha^* = 0.032$.

Type II error: Under $H_a, Y \sim \text{Poisson}(10)$.

$$\beta^* = P(Y < 10 \mid Y \sim \text{Poisson}(10)) = 0.458$$

Example: Poisson (continued)

Finding the critical value: Under $H_0, Y = \sum X_i \sim \text{Poisson}(5)$.

If we use $c = 10$: $\alpha^* = P(Y \geq 10 \mid Y \sim \text{Poisson}(5)) = 1 - 0.968 = 0.032$

If we use $c = 9$: $\alpha^* = P(Y \geq 9 \mid Y \sim \text{Poisson}(5)) = 1 - 0.932 = 0.068$

For $\alpha \leq 0.05$, we use $c = 10$, giving $\alpha^* = 0.032$.

Type II error: Under $H_a, Y \sim \text{Poisson}(10)$.

$$\beta^* = P(Y < 10 \mid Y \sim \text{Poisson}(10)) = 0.458$$

▶ Code

```
alpha* = 0.03182806
```

▶ Code

```
beta*  = 0.4579297
```

▶ Code

```
Power  = 0.5420703
```

Example: Poisson (continued)

The Neyman-Pearson guarantee: Any other test with $\alpha \le 0.032$ must have $\beta \ge 0.458$.

Example: Poisson (continued)

The Neyman-Pearson guarantee: Any other test with $\alpha \leq 0.032$ must have $\beta \geq 0.458$.

The power is only $1 - 0.458 = 0.542$. Why so low?

Example: Poisson (continued)

The Neyman-Pearson guarantee: Any other test with $\alpha \leq 0.032$ must have $\beta \geq 0.458$.

The power is only $1 - 0.458 = 0.542$. Why so low?

Because $n = 5$ **is too small** to discriminate between $\lambda = 1$ and $\lambda = 2$.

For $n = 10$: best test with $\alpha \leq 0.05$ uses $c = 16$, giving $\alpha^* = 0.049$ and $\beta^* = 0.157$ (power = 0.843).

Example: Poisson (continued)

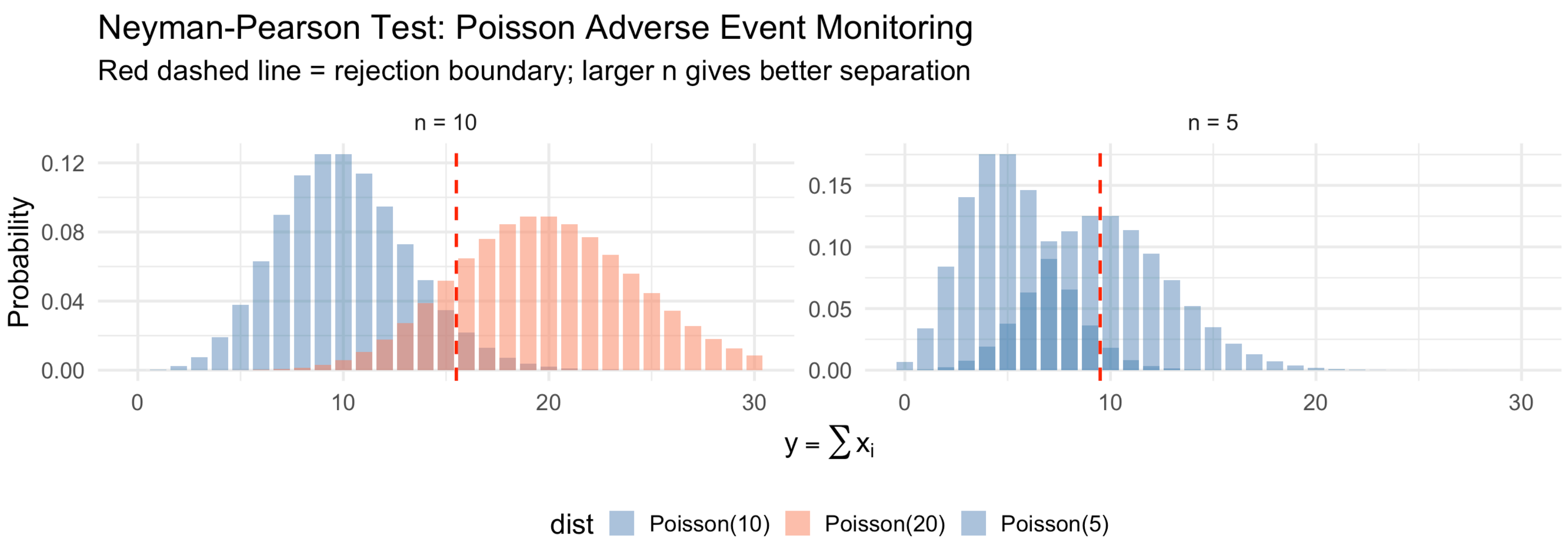
The Neyman-Pearson guarantee: Any other test with $\alpha \leq 0.032$ must have $\beta \geq 0.458$.

The power is only $1 - 0.458 = 0.542$. Why so low?

Because $n = 5$ is too small to discriminate between $\lambda = 1$ and $\lambda = 2$.

For $n = 10$: best test with $\alpha \leq 0.05$ uses $c = 16$, giving $\alpha^* = 0.049$ and $\beta^* = 0.157$ (power = 0.843).

▶ Code



Medical Example: Exponential Survival Times (C&H Example 8.18)

In an oncology trial, patient survival times follow $\mathrm{Exp}(\lambda)$. We want to test whether the hazard rate has increased from a historical baseline.

Test $H_0 : \lambda = 8$ vs $H_a : \lambda = 10$ based on $n = 9$ patients.

Medical Example: Exponential Survival Times (C&H Example 8.18)

In an oncology trial, patient survival times follow $\mathrm{Exp}(\lambda)$. We want to test whether the hazard rate has increased from a historical baseline.

Test $H_0 : \lambda = 8$ vs $H_a : \lambda = 10$ based on $n = 9$ patients.

Likelihood ratio:

$$\frac{L(\lambda = 8)}{L(\lambda = 10)} = \frac{8^9 e^{-8\sum x_i}}{10^9 e^{-10\sum x_i}} = (0.8)^9 e^{2\sum x_i}$$

Medical Example: Exponential Survival Times (C&H Example 8.18)

In an oncology trial, patient survival times follow $\mathrm{Exp}(\lambda)$. We want to test whether the hazard rate has increased from a historical baseline.

Test $H_0 : \lambda = 8$ vs $H_a : \lambda = 10$ based on $n = 9$ patients.

Likelihood ratio:

$$\frac{L(\lambda = 8)}{L(\lambda = 10)} = \frac{8^9 e^{-8\sum x_i}}{10^9 e^{-10\sum x_i}} = (0.8)^9 e^{2\sum x_i}$$

(Note: Chihara uses $L(\theta_0)/L(\theta_a)$, rejecting for small values.)

Reject H_0 when this ratio is small, i.e., when $\sum x_i < c_2$.

Medical Example: Exponential Survival Times (C&H Example 8.18)

In an oncology trial, patient survival times follow $\text{Exp}(\lambda)$. We want to test whether the hazard rate has increased from a historical baseline.

Test $H_0 : \lambda = 8$ vs $H_a : \lambda = 10$ based on $n = 9$ patients.

Likelihood ratio:

$$\frac{L(\lambda = 8)}{L(\lambda = 10)} = \frac{8^9 e^{-8 \sum x_i}}{10^9 e^{-10 \sum x_i}} = (0.8)^9 e^{2 \sum x_i}$$

(Note: Chihara uses $L(\theta_0)/L(\theta_a)$, rejecting for small values.)

Reject H_0 when this ratio is small, i.e., when $\sum x_i < c_2$.

Under H_0: $\sum_{i=1}^{9} X_i \sim \text{Gamma}(9, 8)$. For $\alpha = 0.05$:

```r
c2 <- qgamma(0.05, shape = 9, rate = 8)
cat("Critical value c2 =", round(c2, 4), "\n")
```

```
Critical value c2 = 0.5869
```

By the Neyman-Pearson Lemma, this is the **best possible** test at this significance level.

Power Functions and UMP Tests

The Power Function

Definition (Power Function, Devore 9.5)

Let Ω_0 and Ω_a be two disjoint sets of possible values of θ. For testing $H_0 : \theta \in \Omega_0$ vs $H_a : \theta \in \Omega_a$ with rejection region R, the **power function** is:

$$\pi(\theta') = P_{\theta'}((X_1, \ldots, X_n) \in R)$$

the probability of rejecting H_0 as a function of θ.

The Power Function

Definition (Power Function, Devore 9.5)

Let Ω_0 and Ω_a be two disjoint sets of possible values of θ. For testing $H_0 : \theta \in \Omega_0$ vs $H_a : \theta \in \Omega_a$ with rejection region R, the **power function** is:

$$\pi(\theta') = P_{\theta'}((X_1, \ldots, X_n) \in R)$$

the probability of rejecting H_0 as a function of θ.

Connection to error probabilities:

$$\pi(\theta') = \begin{cases} P(\text{Type I error}) = \alpha(\theta') & \text{when } \theta' \in \Omega_0 \\ 1 - P(\text{Type II error}) = 1 - \beta(\theta') & \text{when } \theta' \in \Omega_a \end{cases}$$

The Power Function

Definition (Power Function, Devore 9.5)

Let Ω_0 and Ω_a be two disjoint sets of possible values of θ. For testing $H_0 : \theta \in \Omega_0$ vs $H_a : \theta \in \Omega_a$ with rejection region R, the **power function** is:

$$\pi(\theta') = P_{\theta'}((X_1, \ldots, X_n) \in R)$$

the probability of rejecting H_0 as a function of θ.

Connection to error probabilities:

$$\pi(\theta') = \begin{cases} P(\text{Type I error}) = \alpha(\theta') & \text{when } \theta' \in \Omega_0 \\ 1 - P(\text{Type II error}) = 1 - \beta(\theta') & \text{when } \theta' \in \Omega_a \end{cases}$$

Ideal power function (unachievable in practice):

$$\pi(\theta') = \begin{cases} 0 & \text{when } \theta' \in \Omega_0 \\ 1 & \text{when } \theta' \in \Omega_a \end{cases}$$

Visualizing the Power Function

Example (Devore 9.25 adapted): Blood pressure reduction (mmHg) with a new antihypertensive follows $N(\mu, 81)$. Test $H_0 : \mu \leq 75$ vs $H_a : \mu < 75$ with $n = 100, \alpha = 0.10$.

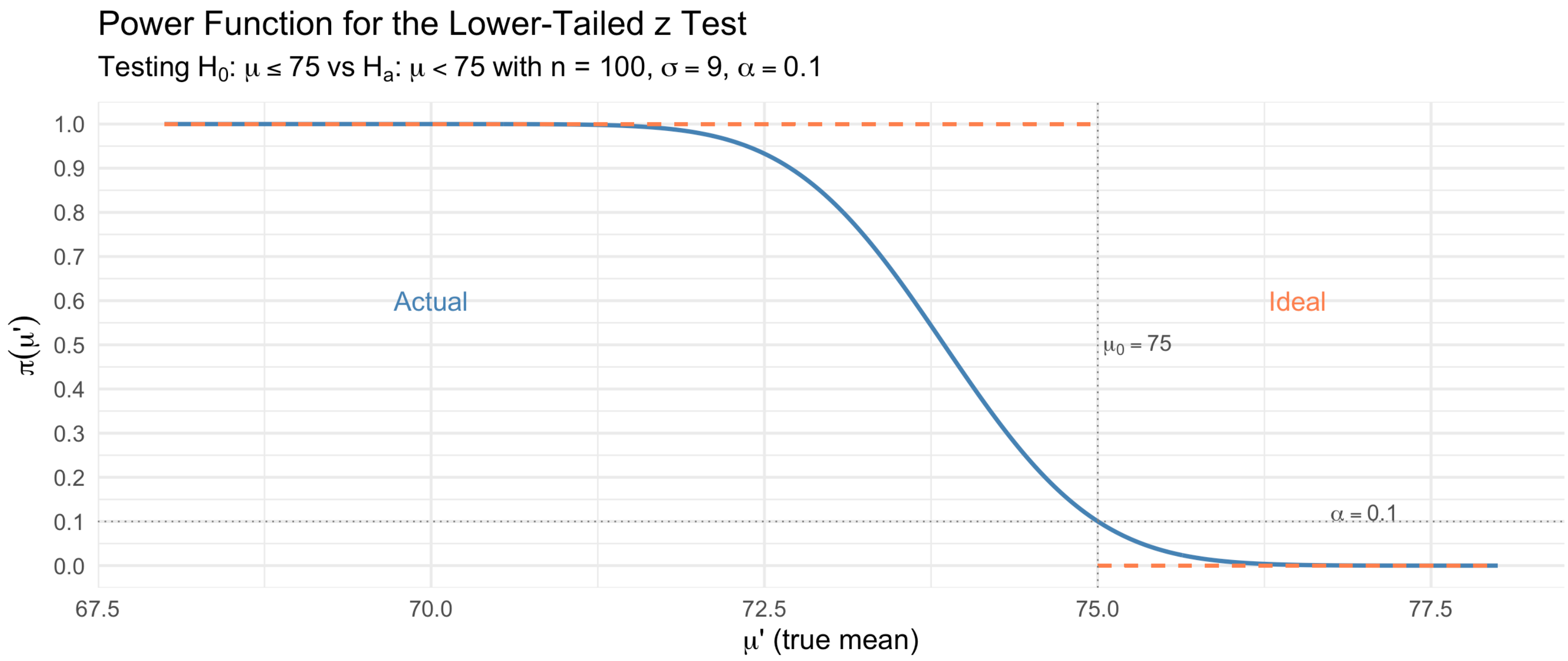
Rejection region: $z \leq -z_{0.10} = -1.28$, i.e., $\bar{x} \leq 73.848$.

Visualizing the Power Function

Example (Devore 9.25 adapted): Blood pressure reduction (mmHg) with a new antihypertensive follows $N(\mu, 81)$. Test $H_0 : \mu \leq 75$ vs $H_a : \mu < 75$ with $n = 100, \alpha = 0.10$.

Rejection region: $z \leq -z_{0.10} = -1.28$, i.e., $\bar{x} \leq 73.848$.

▶ Code



Properties of the Power Function

From the figure, we observe several important properties:

Properties of the Power Function

From the figure, we observe several important properties:

1. **At the boundary** ($\mu = 75$): $\pi(75) = \alpha = 0.10$ by design

Properties of the Power Function

From the figure, we observe several important properties:

1. **At the boundary** ($\mu = 75$): $\pi(75) = \alpha = 0.10$ by design

2. **In** Ω_0 ($\mu > 75$): $\pi(\mu') < 0.10$ – power is below α (good!)

Properties of the Power Function

From the figure, we observe several important properties:

1. **At the boundary** ($\mu = 75$): $\pi(75) = \alpha = 0.10$ by design

2. **In** Ω_0 ($\mu > 75$): $\pi(\mu') < 0.10$ – power is below α (good!)

3. **Near the boundary** (μ slightly below 75): power is barely above α – small departures are hard to detect

Properties of the Power Function

From the figure, we observe several important properties:

1. **At the boundary** ($\mu = 75$): $\pi(75) = \alpha = 0.10$ by design
2. **In Ω_0** ($\mu > 75$): $\pi(\mu') < 0.10$ – power is below α (good!)
3. **Near the boundary** (μ slightly below 75): power is barely above α – small departures are hard to detect
4. **Far from H_0** ($\mu \ll 75$): $\pi(\mu') \to 1$ – large departures are easily detected

Properties of the Power Function

From the figure, we observe several important properties:

1. **At the boundary** ($\mu = 75$): $\pi(75) = \alpha = 0.10$ by design

2. **In** Ω_0 ($\mu > 75$): $\pi(\mu') < 0.10$ – power is below α (good!)

3. **Near the boundary** (μ slightly below 75): power is barely above α – small departures are hard to detect

4. **Far from** H_0 ($\mu \ll 75$): $\pi(\mu') \to 1$ – large departures are easily detected

5. **As** n **increases:** the actual power function approaches the ideal step function

Properties of the Power Function

From the figure, we observe several important properties:

1. **At the boundary** ($\mu = 75$): $\pi(75) = \alpha = 0.10$ by design
2. **In** Ω_0 ($\mu > 75$): $\pi(\mu') < 0.10$ – power is below α (good!)
3. **Near the boundary** (μ slightly below 75): power is barely above α – small departures are hard to detect
4. **Far from** H_0 ($\mu \ll 75$): $\pi(\mu') \to 1$ – large departures are easily detected
5. **As** n **increases:** the actual power function approaches the ideal step function

Clinical Implication

In designing a clinical trial, we need to choose n large enough so that the power function is acceptably close to 1 at the **minimum clinically important difference**. This connects directly to the sample size calculations from Lesson 14.

Uniformly Most Powerful (UMP) Tests

The NP Lemma handles simple vs simple. What about composite alternatives?

Uniformly Most Powerful (UMP) Tests

The NP Lemma handles simple vs simple. What about composite alternatives?

Definition (Uniformly Most Powerful Test)

A test is **uniformly most powerful (UMP)** at level α if it maximizes $\pi(\theta')$ for **every** $\theta' \in \Omega_a$, subject to $\pi(\theta') \leq \alpha$ for all $\theta' \in \Omega_0$.

Uniformly Most Powerful (UMP) Tests

The NP Lemma handles simple vs simple. What about composite alternatives?

Definition (Uniformly Most Powerful Test)

A test is **uniformly most powerful (UMP)** at level α if it maximizes $\pi(\theta')$ for **every** $\theta' \in \Omega_a$, subject to $\pi(\theta') \leq \alpha$ for all $\theta' \in \Omega_0$.

When do UMP tests exist? If the NP rejection region **does not depend on the specific alternative** θ_a, then the same test is most powerful for *every* $\theta_a \in \Omega_a$.

UMP Example: Poisson (Devore Example 9.26)

Returning to the Poisson adverse event example with $n = 5$:

Test $H_0 : \lambda \leq 1$ vs $H_a : \lambda > 1$.

UMP Example: Poisson (Devore Example 9.26)

Returning to the Poisson adverse event example with $n = 5$:

Test $H_0 : \lambda \leq 1$ vs $H_a : \lambda > 1$.

For **any** specific $\lambda_a > 1$, the NP test of $H_0 : \lambda = 1$ vs $H_a : \lambda = \lambda_a$ rejects when $\sum x_i \geq c$.

UMP Example: Poisson (Devore Example 9.26)

Returning to the Poisson adverse event example with $n = 5$:

Test $H_0 : \lambda \leq 1$ vs $H_a : \lambda > 1$.

For **any** specific $\lambda_a > 1$, the NP test of $H_0 : \lambda = 1$ vs $H_a : \lambda = \lambda_a$ rejects when $\sum x_i \geq c$.

Key observation: The rejection region $\sum x_i \geq 10$ is the **same** regardless of which $\lambda_a > 1$ we choose! (The critical value c is determined by H_0 alone.)

UMP Example: Poisson (Devore Example 9.26)

Returning to the Poisson adverse event example with $n = 5$:

Test $H_0 : \lambda \leq 1$ vs $H_a : \lambda > 1$.

For **any** specific $\lambda_a > 1$, the NP test of $H_0 : \lambda = 1$ vs $H_a : \lambda = \lambda_a$ rejects when $\sum x_i \geq c$.

Key observation: The rejection region $\sum x_i \geq 10$ is the **same** regardless of which $\lambda_a > 1$ we choose! (The critical value c is determined by H_0 alone.)

Therefore, this test is **UMP** at level $\alpha^* = 0.032$.

UMP Example: Poisson (Devore Example 9.26)

Returning to the Poisson adverse event example with $n = 5$:

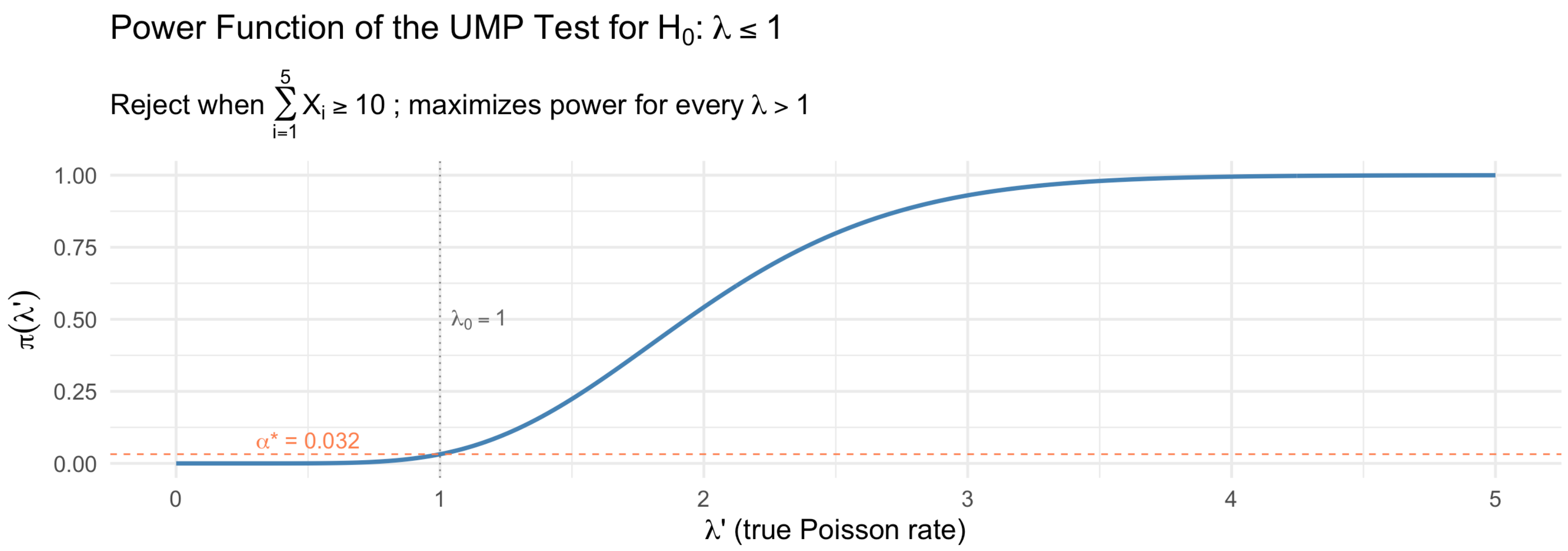
Test $H_0 : \lambda \leq 1$ vs $H_a : \lambda > 1$.

For **any** specific $\lambda_a > 1$, the NP test of $H_0 : \lambda = 1$ vs $H_a : \lambda = \lambda_a$ rejects when $\sum x_i \geq c$.

Key observation: The rejection region $\sum x_i \geq 10$ is the **same** regardless of which $\lambda_a > 1$ we choose! (The critical value c is determined by H_0 alone.)

Therefore, this test is **UMP** at level $\alpha^* = 0.032$.

▶ Code



When UMP Tests Don't Exist

Important Limitation

UMP tests are **fairly rare**, especially in the commonly encountered situation of testing a two-sided alternative.

When UMP Tests Don't Exist

> ⚠ **Important Limitation**
>
> UMP tests are **fairly rare**, especially in the commonly encountered situation of testing a two-sided alternative.

Example: Testing $H_0 : \mu = \mu_0$ vs $H_a : \mu \neq \mu_0$ with $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$, σ known.

- For $\mu_a > \mu_0$: NP test rejects when $\bar{x}$ is **large** (upper tail)
- For $\mu_a < \mu_0$: NP test rejects when $\bar{x}$ is **small** (lower tail)
- These are **different rejection regions** — no single test is best for all $\mu_a \neq \mu_0$

When UMP Tests Don't Exist

> ⚠ **Important Limitation**
>
> UMP tests are **fairly rare**, especially in the commonly encountered situation of testing a two-sided alternative.

Example: Testing $H_0 : \mu = \mu_0$ vs $H_a : \mu \neq \mu_0$ with $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$, σ known.

- For $\mu_a > \mu_0$: NP test rejects when $\bar{x}$ is **large** (upper tail)
- For $\mu_a < \mu_0$: NP test rejects when $\bar{x}$ is **small** (lower tail)
- These are **different rejection regions** — no single test is best for all $\mu_a \neq \mu_0$

Resolution: Restrict attention to **unbiased tests** (tests where $\pi(\theta') \geq \alpha$ for all $\theta' \in \Omega_a$). Under this restriction:

- The two-sided z test is **UMP unbiased** when σ is known
- The one-sample t test is **UMP unbiased** when σ is unknown

Unbiased Tests

Definition (Unbiased Test)

A test is **unbiased** if the power under any alternative is at least as large as the power under any null value:

$$\inf_{\theta' \in \Omega_a} \pi(\theta') \geq \sup_{\theta' \in \Omega_0} \pi(\theta')$$

Equivalently: we are at least as likely to reject H_0 when it is false as when it is true.

Unbiased Tests

Definition (Unbiased Test)

A test is **unbiased** if the power under any alternative is at least as large as the power under any null value:

$$\inf_{\theta' \in \Omega_a} \pi(\theta') \geq \sup_{\theta' \in \Omega_0} \pi(\theta')$$

Equivalently: we are at least as likely to reject H_0 when it is false as when it is true.

Why is this desirable? A biased test would reject H_0 *less* often for some alternatives than it does when H_0 is true — that would defeat the purpose of the test!

Unbiased Tests

Definition (Unbiased Test)

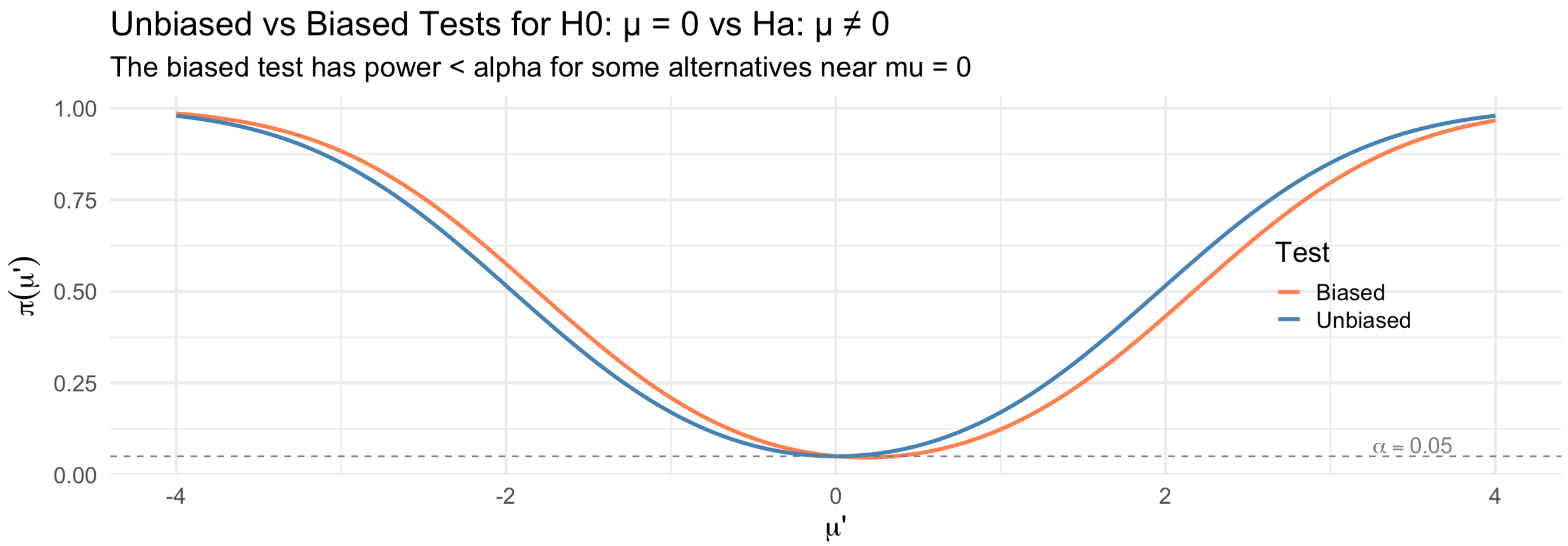
A test is **unbiased** if the power under any alternative is at least as large as the power under any null value:

$$\inf_{\theta' \in \Omega_a} \pi(\theta') \geq \sup_{\theta' \in \Omega_0} \pi(\theta')$$

Equivalently: we are at least as likely to reject H_0 when it is false as when it is true.

Why is this desirable? A biased test would reject H_0 *less* often for some alternatives than it does when H_0 is true — that would defeat the purpose of the test!

▶ Code



Likelihood Ratio Tests (Devore 9.5)

From Simple to Composite Hypotheses

The NP Lemma handles simple vs simple hypotheses. But in practice, we need tests for:

$$H_0 : \theta \in \Omega_0 \quad \text{vs} \quad H_a : \theta \in \Omega_a$$

where one or both hypotheses may be composite.

From Simple to Composite Hypotheses

The NP Lemma handles simple vs simple hypotheses. But in practice, we need tests for:

$$H_0 : \theta \in \Omega_0 \quad \text{vs} \quad H_a : \theta \in \Omega_a$$

where one or both hypotheses may be composite.

The likelihood ratio principle provides a **general recipe** for constructing test statistics in virtually any testing situation.

From Simple to Composite Hypotheses

The NP Lemma handles simple vs simple hypotheses. But in practice, we need tests for:

$$H_0 : \theta \in \Omega_0 \quad \text{vs} \quad H_a : \theta \in \Omega_a$$

where one or both hypotheses may be composite.

The likelihood ratio principle provides a **general recipe** for constructing test statistics in virtually any testing situation.

The Likelihood Ratio Test (LRT) Procedure

1. Find $\hat{\theta}_{mle}$, the MLE of θ over the **full** parameter space $\Omega = \Omega_0 \cup \Omega_a$
2. Find $\hat{\theta}_0$, the **restricted MLE** of θ over Ω_0 only
3. Form the **likelihood ratio statistic**:

$$\Lambda = \frac{L(\hat{\theta}_0)}{L(\hat{\theta}_{mle})}$$

4. Reject H_0 when $\Lambda \leq k$ (i.e., when the restricted MLE fits much worse than the unrestricted MLE)

Understanding the LRT Statistic

$$\Lambda = \frac{L(\hat{\theta}_0)}{L(\hat{\theta}_{mle})} = \frac{\max_{\theta \in \Omega_0} L(\theta)}{\max_{\theta \in \Omega} L(\theta)}$$

Understanding the LRT Statistic

$$\Lambda = \frac{L(\hat{\theta}_0)}{L(\hat{\theta}_{mle})} = \frac{\max_{\theta \in \Omega_0} L(\theta)}{\max_{\theta \in \Omega} L(\theta)}$$

Key properties:

- Always $0 \leq \Lambda \leq 1$ (because $\Omega_0 \subseteq \Omega$, so the restricted maximum $\leq$ unrestricted maximum)
- $\Lambda \approx 1$: the data are about equally consistent with H_0 and the overall best fit
- $\Lambda \approx 0$: the data are **much** more consistent with H_a than with H_0

Understanding the LRT Statistic

$$\Lambda = \frac{L(\hat{\theta}_0)}{L(\hat{\theta}_{mle})} = \frac{\max_{\theta \in \Omega_0} L(\theta)}{\max_{\theta \in \Omega} L(\theta)}$$

Key properties:

- Always $0 \leq \Lambda \leq 1$ (because $\Omega_0 \subseteq \Omega$, so the restricted maximum $\leq$ unrestricted maximum)
- $\Lambda \approx 1$: the data are about equally consistent with H_0 and the overall best fit
- $\Lambda \approx 0$: the data are **much** more consistent with H_a than with H_0

Connection to NP Lemma: When both hypotheses are simple ($\Omega_0 = \{\theta_0\}$ and $\Omega_a = \{\theta_a\}$):

$$\Lambda = \frac{L(\theta_0)}{\max\{L(\theta_0), L(\theta_a)\}} = \begin{cases} L(\theta_0)/L(\theta_a) & \text{if } L(\theta_a) \geq L(\theta_0) \\ 1 & \text{otherwise} \end{cases}$$

This is the reciprocal of the NP statistic (when H_a is more likely), so rejecting for small Λ is equivalent to rejecting for large Λ^*.

LRT Example 1: Normal Mean, Known Variance (Devore Example 9.27 style)

Systolic blood pressure readings follow $N(\mu, 100)$ (known $\sigma = 10$). A clinical trial tests whether a new treatment changes the population mean from a target value.

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_a : \mu \neq \mu_0$$

LRT Example 1: Normal Mean, Known Variance (Devore Example 9.27 style)

Systolic blood pressure readings follow $N(\mu, 100)$ (known $\sigma = 10$). A clinical trial tests whether a new treatment changes the population mean from a target value.

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_a : \mu \neq \mu_0$$

Step 1: Unrestricted MLE. Over $\Omega = \mathbb{R}$: $\hat{\mu}_{mle} = \bar{x}$

LRT Example 1: Normal Mean, Known Variance (Devore Example 9.27 style)

Systolic blood pressure readings follow $N(\mu, 100)$ (known $\sigma = 10$). A clinical trial tests whether a new treatment changes the population mean from a target value.

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_a : \mu \neq \mu_0$$

Step 1: Unrestricted MLE. Over $\Omega = \mathbb{R}$: $\hat{\mu}_{mle} = \bar{x}$

Step 2: Restricted MLE. Over $\Omega_0 = \{\mu_0\}$: $\hat{\mu}_0 = \mu_0$

LRT Example 1: Normal Mean, Known Variance (Devore Example 9.27 style)

Systolic blood pressure readings follow $N(\mu, 100)$ (known $\sigma = 10$). A clinical trial tests whether a new treatment changes the population mean from a target value.

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_a : \mu \neq \mu_0$$

Step 1: Unrestricted MLE. Over $\Omega = \mathbb{R}$: $\hat{\mu}_{mle} = \bar{x}$

Step 2: Restricted MLE. Over $\Omega_0 = \{\mu_0\}$: $\hat{\mu}_0 = \mu_0$

Step 3: Likelihood ratio.

$$\Lambda = \frac{L(\mu_0)}{L(\bar{x})} = \frac{\prod \frac{1}{\sqrt{2\pi}\sigma} e^{-(x_i-\mu_0)^2/(2\sigma^2)}}{\prod \frac{1}{\sqrt{2\pi}\sigma} e^{-(x_i-\bar{x})^2/(2\sigma^2)}} = e^{-\frac{1}{2\sigma^2}\left[\sum(x_i-\mu_0)^2-\sum(x_i-\bar{x})^2\right]}$$

LRT Example 1: Normal Mean, Known Variance (Devore Example 9.27 style)

Systolic blood pressure readings follow $N(\mu, 100)$ (known $\sigma = 10$). A clinical trial tests whether a new treatment changes the population mean from a target value.

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_a : \mu \neq \mu_0$$

Step 1: Unrestricted MLE. Over $\Omega = \mathbb{R}$: $\hat{\mu}_{mle} = \bar{x}$

Step 2: Restricted MLE. Over $\Omega_0 = \{\mu_0\}$: $\hat{\mu}_0 = \mu_0$

Step 3: Likelihood ratio.

$$\Lambda = \frac{L(\mu_0)}{L(\bar{x})} = \frac{\prod \frac{1}{\sqrt{2\pi}\sigma} e^{-(x_i-\mu_0)^2/(2\sigma^2)}}{\prod \frac{1}{\sqrt{2\pi}\sigma} e^{-(x_i-\bar{x})^2/(2\sigma^2)}} = e^{-\frac{1}{2\sigma^2}\left[\sum(x_i-\mu_0)^2 - \sum(x_i-\bar{x})^2\right]}$$

Since $\sum(x_i - \mu_0)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - \mu_0)^2$:

$$\Lambda = e^{-n(\bar{x}-\mu_0)^2/(2\sigma^2)}$$

LRT $\Rightarrow$ z Test

Reject when $\Lambda \leq k$:

$$e^{-n(\bar{x}-\mu_0)^2/(2\sigma^2)} \leq k$$

LRT $\Rightarrow$ z Test

Reject when $\Lambda \leq k$:

$$e^{-n(\bar{x}-\mu_0)^2/(2\sigma^2)} \leq k$$

Taking $-\ln$ of both sides:

$$\frac{n(\bar{x}-\mu_0)^2}{2\sigma^2} \geq -\ln(k) = c'$$

LRT $\Rightarrow$ z Test

Reject when $\Lambda \le k$:

$$e^{-n(\bar{x}-\mu_0)^2/(2\sigma^2)} \le k$$

Taking $-\ln$ of both sides:

$$\frac{n(\bar{x}-\mu_0)^2}{2\sigma^2} \ge -\ln(k) = c'$$

$$\left|\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}\right| \ge c$$

LRT $\Rightarrow$ z Test

Reject when $\Lambda \leq k$:

$$e^{-n(\bar{x}-\mu_0)^2/(2\sigma^2)} \leq k$$

Taking $-\ln$ of both sides:

$$\frac{n(\bar{x}-\mu_0)^2}{2\sigma^2} \geq -\ln(k) = c'$$

$$\left|\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}\right| \geq c$$

This is the **two-sided z test**! Setting $c = z_{\alpha/2}$ gives the test at level α.

LRT $\Rightarrow$ z Test

Reject when $\Lambda \leq k$:

$$e^{-n(\bar{x}-\mu_0)^2/(2\sigma^2)} \leq k$$

Taking $-\ln$ of both sides:

$$\frac{n(\bar{x}-\mu_0)^2}{2\sigma^2} \geq -\ln(k) = c'$$

$$\left|\frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}\right| \geq c$$

This is the **two-sided z test**! Setting $c = z_{\alpha/2}$ gives the test at level α.

Key Result

The **likelihood ratio test** for $H_0 : \mu = \mu_0$ vs $H_a : \mu \neq \mu_0$ when σ is known is exactly the **two-sided z test**. The LRT principle automatically "discovers" the z test.

LRT Example 2: Normal Mean, Unknown Variance (Devore Example 9.27)

Now suppose $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ **unknown**.

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_a : \mu \neq \mu_0$$

LRT Example 2: Normal Mean, Unknown Variance (Devore Example 9.27)

Now suppose $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ **unknown**.

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_a : \mu \neq \mu_0$$

Step 1: Unrestricted MLE over $\Omega = \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}$:

$$\hat{\mu}_{mle} = \bar{x}, \quad \hat{\sigma}^2_{mle} = \frac{1}{n} \sum (x_i - \bar{x})^2$$

LRT Example 2: Normal Mean, Unknown Variance (Devore Example 9.27)

Now suppose $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ **unknown**.

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_a : \mu \neq \mu_0$$

Step 1: Unrestricted MLE over $\Omega = \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}$:

$$\hat{\mu}_{mle} = \bar{x}, \quad \hat{\sigma}^2_{mle} = \frac{1}{n} \sum (x_i - \bar{x})^2$$

Step 2: Restricted MLE over $\Omega_0 = \{(\mu_0, \sigma^2) : \sigma^2 > 0\}$:

$$\hat{\mu}_0 = \mu_0, \quad \hat{\sigma}^2_0 = \frac{1}{n} \sum (x_i - \mu_0)^2$$

LRT Example 2: Normal Mean, Unknown Variance (Devore Example 9.27)

Now suppose $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ **unknown**.

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_a : \mu \neq \mu_0$$

Step 1: Unrestricted MLE over $\Omega = \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}$:

$$\hat{\mu}_{mle} = \bar{x}, \quad \hat{\sigma}^2_{mle} = \frac{1}{n} \sum (x_i - \bar{x})^2$$

Step 2: Restricted MLE over $\Omega_0 = \{(\mu_0, \sigma^2) : \sigma^2 > 0\}$:

$$\hat{\mu}_0 = \mu_0, \quad \hat{\sigma}^2_0 = \frac{1}{n} \sum (x_i - \mu_0)^2$$

Note: under H_0, we still estimate σ^2 by MLE, but with μ fixed at μ_0.

LRT $\Rightarrow$ t Test (Devore Example 9.27)

Step 3: Likelihood ratio. After substitution and simplification:

$$\Lambda = \left[\frac{\hat{\sigma}^2_{mle}}{\hat{\sigma}^2_0} \right]^{n/2} = \left[\frac{\sum (x_i - \bar{x})^2}{\sum (x_i - \mu_0)^2} \right]^{n/2}$$

LRT $\Rightarrow$ t Test (Devore Example 9.27)

Step 3: Likelihood ratio. After substitution and simplification:

$$\Lambda = \left[\frac{\hat{\sigma}^2_{mle}}{\hat{\sigma}^2_0}\right]^{n/2} = \left[\frac{\sum(x_i - \bar{x})^2}{\sum(x_i - \mu_0)^2}\right]^{n/2}$$

Since $\sum(x_i - \mu_0)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - \mu_0)^2$:

$$\Lambda = \left[\frac{1}{1 + \frac{n(\bar{x}-\mu_0)^2}{\sum(x_i-\bar{x})^2}}\right]^{n/2} = \left[\frac{1}{1 + t^2/(n-1)}\right]^{n/2}$$

where $t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$ is the **t statistic**.

LRT $\Rightarrow$ t Test (Devore Example 9.27)

Step 3: Likelihood ratio. After substitution and simplification:

$$\Lambda = \left[\frac{\hat{\sigma}^2_{mle}}{\hat{\sigma}^2_0}\right]^{n/2} = \left[\frac{\sum(x_i - \bar{x})^2}{\sum(x_i - \mu_0)^2}\right]^{n/2}$$

Since $\sum(x_i - \mu_0)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - \mu_0)^2$:

$$\Lambda = \left[\frac{1}{1 + \frac{n(\bar{x}-\mu_0)^2}{\sum(x_i-\bar{x})^2}}\right]^{n/2} = \left[\frac{1}{1 + t^2/(n-1)}\right]^{n/2}$$

where $t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$ is the **t statistic**.

Reject when Λ is small $\iff t^2/(n-1)$ is large $\iff |t| \geq c$.

Setting $c = t_{\alpha/2,n-1}$ gives the **one-sample t test**.

LRT $\Rightarrow$ t Test (Devore Example 9.27)

Step 3: Likelihood ratio. After substitution and simplification:

$$\Lambda = \left[\frac{\hat{\sigma}^2_{mle}}{\hat{\sigma}^2_0}\right]^{n/2} = \left[\frac{\sum(x_i - \bar{x})^2}{\sum(x_i - \mu_0)^2}\right]^{n/2}$$

Since $\sum(x_i - \mu_0)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - \mu_0)^2$:

$$\Lambda = \left[\frac{1}{1 + \frac{n(\bar{x}-\mu_0)^2}{\sum(x_i-\bar{x})^2}}\right]^{n/2} = \left[\frac{1}{1 + t^2/(n-1)}\right]^{n/2}$$

where $t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$ is the **t statistic**.

Reject when Λ is small $\iff t^2/(n-1)$ is large $\iff |t| \geq c$.

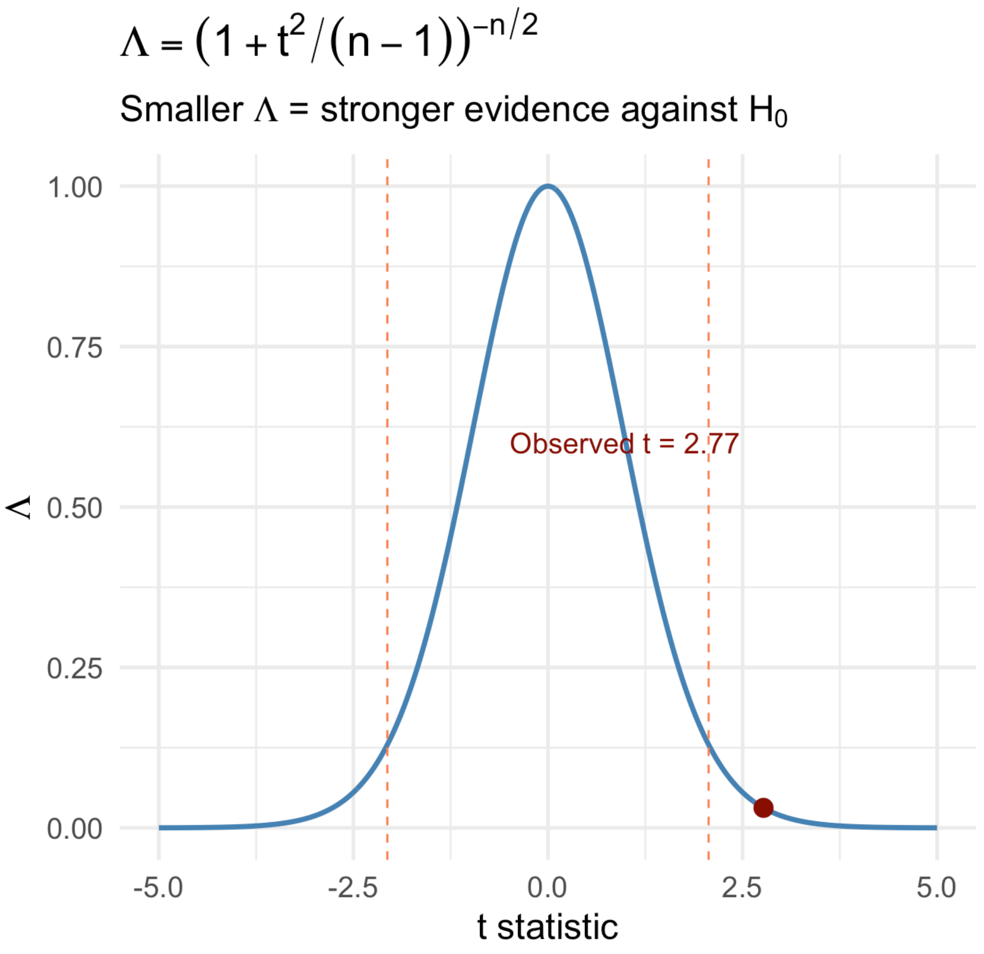
Setting $c = t_{\alpha/2,n-1}$ gives the **one-sample t test**.

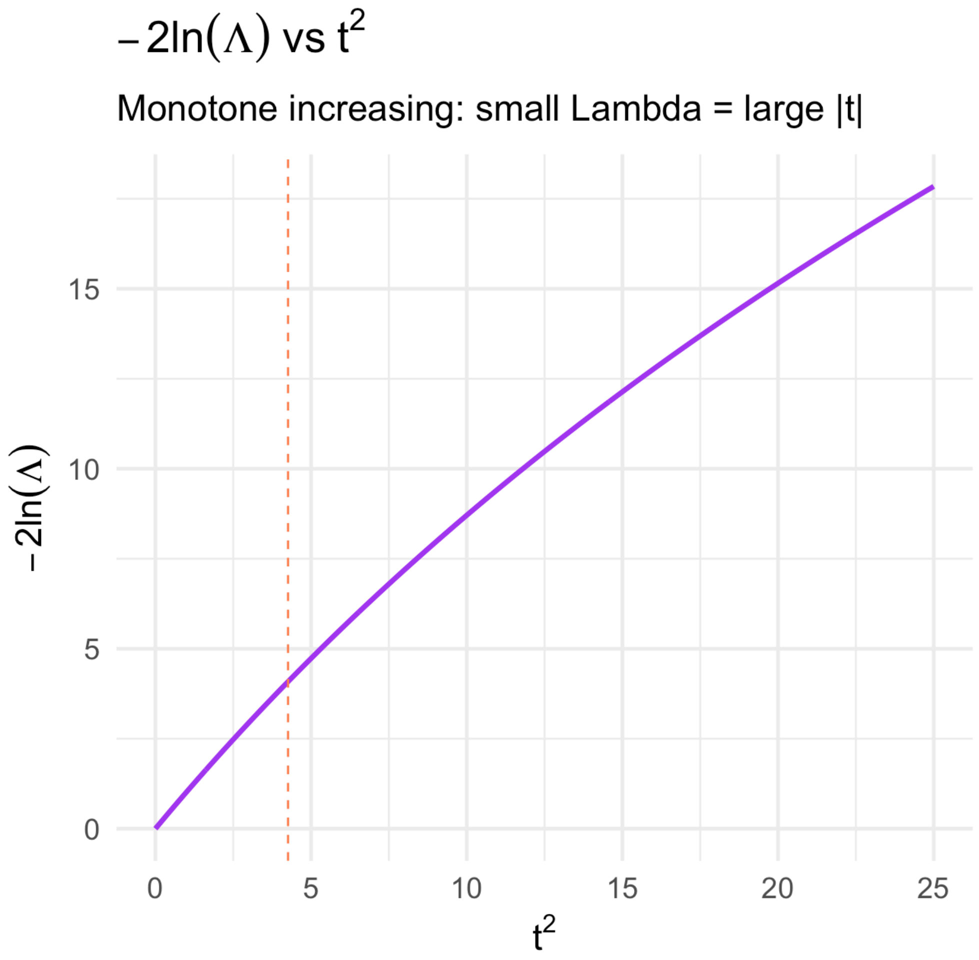
Key Result

The **likelihood ratio test** for $H_0 : \mu = \mu_0$ vs $H_a : \mu \neq \mu_0$ when σ is unknown is exactly the **one-sample t test**. Many classical tests can be derived from the LRT principle.

LRT and the t Test: Simulation Illustration

▶ Code





Medical Application: Drug Efficacy Trial

A Phase III clinical trial tests whether a new cholesterol-lowering drug changes mean LDL from baseline $\mu_0 = 130$ mg/dL. Data from $n = 30$ patients:

```r
set.seed(42)
n <- 30
mu_0 <- 130
ldl_data <- rnorm(n, mean = 122, sd = 18)

x_bar <- mean(ldl_data)
s_val <- sd(ldl_data)
t_obs <- (x_bar - mu_0) / (s_val / sqrt(n))

cat("Sample mean:", round(x_bar, 2), "mg/dL\n")
```

```
Sample mean: 123.23 mg/dL
```

```r
cat("Sample SD:  ", round(s_val, 2), "mg/dL\n")
```

```
Sample SD:   22.59 mg/dL
```

```r
cat("t statistic:", round(t_obs, 3), "\n")
```

```
t statistic: -1.64
```

Medical Application: Drug Efficacy Trial

A Phase III clinical trial tests whether a new cholesterol-lowering drug changes mean LDL from baseline $\mu_0 = 130$ mg/dL. Data from $n = 30$ patients:

```
set.seed(42)
n <- 30
mu_0 <- 130
ldl_data <- rnorm(n, mean = 122, sd = 18)

x_bar <- mean(ldl_data)
s_val <- sd(ldl_data)
t_obs <- (x_bar - mu_0) / (s_val / sqrt(n))

cat("Sample mean:", round(x_bar, 2), "mg/dL\n")
```

```
Sample mean: 123.23 mg/dL
```

```
cat("Sample SD:  ", round(s_val, 2), "mg/dL\n")
```

```
Sample SD:   22.59 mg/dL
```

```
cat("t statistic:", round(t_obs, 3), "\n")
```

```
t statistic: -1.64
```

```
# LRT statistic
Lambda <- (1 / (1 + t_obs^2 / (n - 1)))^(n / 2)
neg2lnLambda <- -2 * log(Lambda)
```

Medical Application: Drug Efficacy Trial

A Phase III clinical trial tests whether a new cholesterol-lowering drug changes mean LDL from baseline $\mu_0 = 130$ mg/dL. Data from $n = 30$ patients:

```
set.seed(42)
n <- 30
mu_0 <- 130
ldl_data <- rnorm(n, mean = 122, sd = 18)

x_bar <- mean(ldl_data)
s_val <- sd(ldl_data)
t_obs <- (x_bar - mu_0) / (s_val / sqrt(n))

cat("Sample mean:", round(x_bar, 2), "mg/dL\n")
```

```
Sample mean: 123.23 mg/dL
```

```
cat("Sample SD:  ", round(s_val, 2), "mg/dL\n")
```

```
Sample SD:   22.59 mg/dL
```

```
cat("t statistic:", round(t_obs, 3), "\n")
```

```
t statistic: -1.64
```

```
# LRT statistic
Lambda <- (1 / (1 + t_obs^2 / (n - 1)))^(n / 2)
neg2lnLambda <- -2 * log(Lambda)
```

Large-Sample LRT Approximation

Wilks' Theorem: The Chi-Squared Approximation

In many situations, the LRT rejection region $\Lambda \leq k$ cannot be expressed in terms of a simple, familiar statistic. A powerful large-sample result resolves this.

Wilks' Theorem: The Chi-Squared Approximation

In many situations, the LRT rejection region $\Lambda \leq k$ cannot be expressed in terms of a simple, familiar statistic. A powerful large-sample result resolves this.

Theorem (Wilks' Theorem, Devore 9.5)

If the sample size n is sufficiently large, then under H_0:

$$-2\ln(\Lambda) \xrightarrow{d} \chi^2_m$$

where m is the difference between the number of free parameters in Ω and the number of free parameters in Ω_0.

Wilks' Theorem: The Chi-Squared Approximation

In many situations, the LRT rejection region $\Lambda \leq k$ cannot be expressed in terms of a simple, familiar statistic. A powerful large-sample result resolves this.

Theorem (Wilks' Theorem, Devore 9.5)

If the sample size n is sufficiently large, then under H_0:

$$-2\ln(\Lambda) \xrightarrow{d} \chi^2_m$$

where m is the difference between the number of free parameters in Ω and the number of free parameters in Ω_0.

How to use it: Since $0 \leq \Lambda \leq 1$, we have $-2\ln(\Lambda) \geq 0$. Reject H_0 when:

$$-2\ln(\Lambda) \geq \chi^2_{\alpha,m}$$

Wilks' Theorem: The Chi-Squared Approximation

In many situations, the LRT rejection region $\Lambda \leq k$ cannot be expressed in terms of a simple, familiar statistic. A powerful large-sample result resolves this.

Theorem (Wilks' Theorem, Devore 9.5)

If the sample size n is sufficiently large, then under H_0:

$$-2\ln(\Lambda) \xrightarrow{d} \chi^2_m$$

where m is the difference between the number of free parameters in Ω and the number of free parameters in Ω_0.

How to use it: Since $0 \leq \Lambda \leq 1$, we have $-2\ln(\Lambda) \geq 0$. Reject H_0 when:

$$-2\ln(\Lambda) \geq \chi^2_{\alpha,m}$$

The degrees of freedom m:

- $m = \dim(\Omega) - \dim(\Omega_0)$ = (number of parameters estimated under H_a) minus (number estimated under H_0)
- Represents the number of "constraints" imposed by H_0

Computing Degrees of Freedom: Examples

Setting	Ω	Ω_0	m
$N(\mu, \sigma^2)$, σ known, $H_0 : \mu = \mu_0$	1 (μ)	0	1
$N(\mu, \sigma^2)$, σ unknown, $H_0 : \mu = \mu_0$	2 (μ, σ^2)	1 (σ^2)	1
Bivariate normal, $H_0 : \mu_1 = \mu_2$ and $\sigma_1 = \sigma_2$	5 ($\mu_1, \mu_2, \sigma_1, \sigma_2, \rho$)	3 (μ, σ, ρ)	2
Multinomial with k categories, $H_0 : p_i = p_{i,0}$	$k - 1$	0	$k - 1$

Computing Degrees of Freedom: Examples

Setting	Ω	Ω_0	m
$N(\mu, \sigma^2)$, σ known, $H_0 : \mu = \mu_0$	1 (μ)	0	1
$N(\mu, \sigma^2)$, σ unknown, $H_0 : \mu = \mu_0$	2 (μ, σ^2)	1 (σ^2)	1
Bivariate normal, $H_0 : \mu_1 = \mu_2$ and $\sigma_1 = \sigma_2$	5 ($\mu_1, \mu_2, \sigma_1, \sigma_2, \rho$)	3 (μ, σ, ρ)	2
Multinomial with k categories, $H_0 : p_i = p_{i,0}$	$k - 1$	0	$k - 1$

Practical Rule

Count the number of parameters that are "free" (need to be estimated) under each hypothesis. The difference is m.

Large-Sample LRT Example: Double Exponential (Devore 9.5)

The Laplace (double exponential) distribution with pdf $f(x;\theta) = \frac{1}{2}e^{-|x-\theta|}$ is sometimes used to model measurement errors. Test $H_0 : \theta = 0$ vs $H_a : \theta \neq 0$.

Large-Sample LRT Example: Double Exponential (Devore 9.5)

The Laplace (double exponential) distribution with pdf $f(x;\theta) = \frac{1}{2}e^{-|x-\theta|}$ is sometimes used to model measurement errors. Test $H_0 : \theta = 0$ vs $H_a : \theta \neq 0$.

Under Ω: The MLE of θ is the **sample median** $\tilde{x}$ (minimizes $\sum |x_i - \theta|$).

Under Ω_0: $\theta = 0$ (no estimation needed).

Large-Sample LRT Example: Double Exponential (Devore 9.5)

The Laplace (double exponential) distribution with pdf $f(x;\theta) = \frac{1}{2}e^{-|x-\theta|}$ is sometimes used to model measurement errors. Test $H_0 : \theta = 0$ vs $H_a : \theta \neq 0$.

Under Ω: The MLE of θ is the **sample median** $\tilde{x}$ (minimizes $\sum |x_i - \theta|$).

Under Ω_0: $\theta = 0$ (no estimation needed).

$$\Lambda = \frac{(1/2)^n e^{-\sum |x_i|}}{(1/2)^n e^{-\sum |x_i - \tilde{x}|}} = e^{-(\sum |x_i| - \sum |x_i - \tilde{x}|)}$$

Large-Sample LRT Example: Double Exponential (Devore 9.5)

The Laplace (double exponential) distribution with pdf $f(x;\theta) = \frac{1}{2}e^{-|x-\theta|}$ is sometimes used to model measurement errors. Test $H_0 : \theta = 0$ vs $H_a : \theta \neq 0$.

Under Ω: The MLE of θ is the **sample median** $\tilde{x}$ (minimizes $\sum |x_i - \theta|$).

Under Ω_0: $\theta = 0$ (no estimation needed).

$$\Lambda = \frac{(1/2)^n e^{-\sum |x_i|}}{(1/2)^n e^{-\sum |x_i - \tilde{x}|}} = e^{-(\sum |x_i| - \sum |x_i - \tilde{x}|)}$$

$$-2\ln(\Lambda) = 2\left(\sum |x_i| - \sum |x_i - \tilde{x}|\right)$$

Large-Sample LRT Example: Double Exponential (Devore 9.5)

The Laplace (double exponential) distribution with pdf $f(x;\theta) = \frac{1}{2}e^{-|x-\theta|}$ is sometimes used to model measurement errors. Test $H_0 : \theta = 0$ vs $H_a : \theta \neq 0$.

Under Ω: The MLE of θ is the **sample median** $\tilde{x}$ (minimizes $\sum |x_i - \theta|$).

Under Ω_0: $\theta = 0$ (no estimation needed).

$$\Lambda = \frac{(1/2)^n e^{-\sum |x_i|}}{(1/2)^n e^{-\sum |x_i - \tilde{x}|}} = e^{-(\sum |x_i| - \sum |x_i - \tilde{x}|)}$$

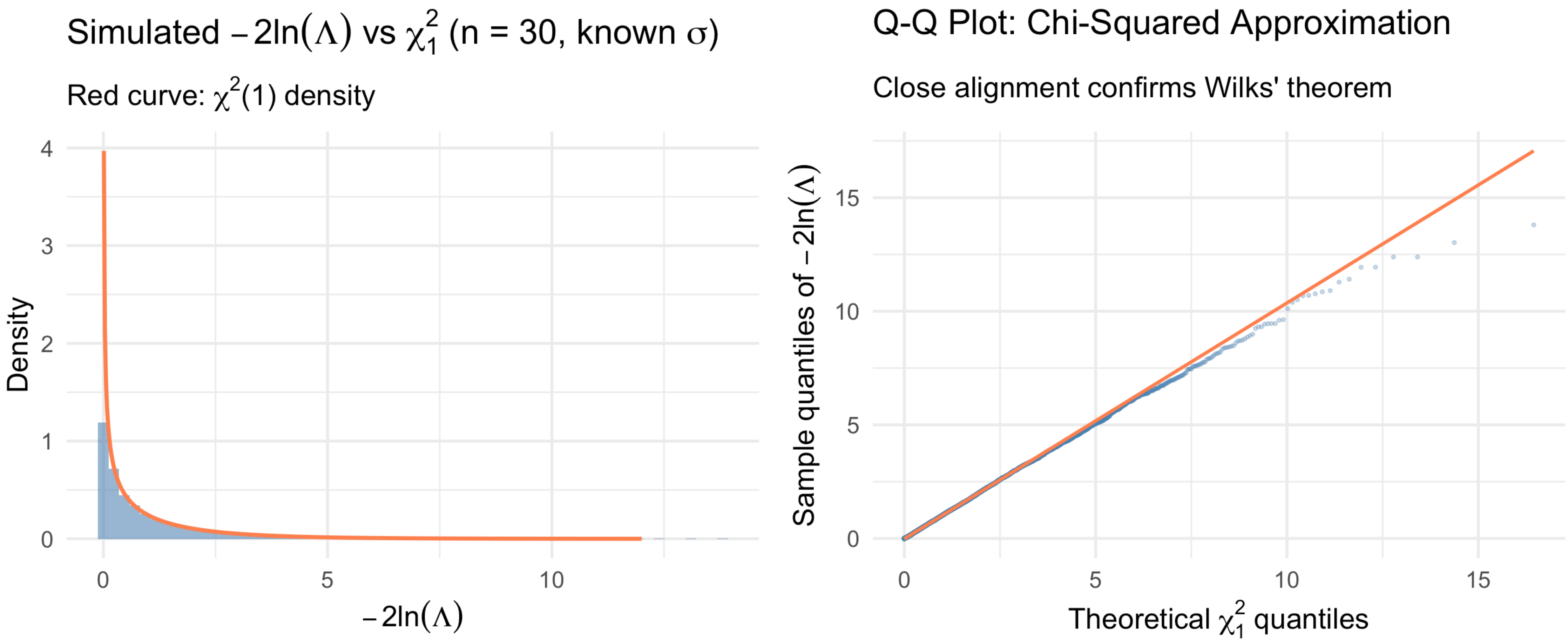
$$-2\ln(\Lambda) = 2\left(\sum |x_i| - \sum |x_i - \tilde{x}|\right)$$

Degrees of freedom: $m = 1 - 0 = 1$, so $-2\ln(\Lambda) \mathrel{\dot\sim} \chi^2_1$ under H_0.

Reject H_0 when $-2\ln(\Lambda) \geq \chi^2_{\alpha,1}$.

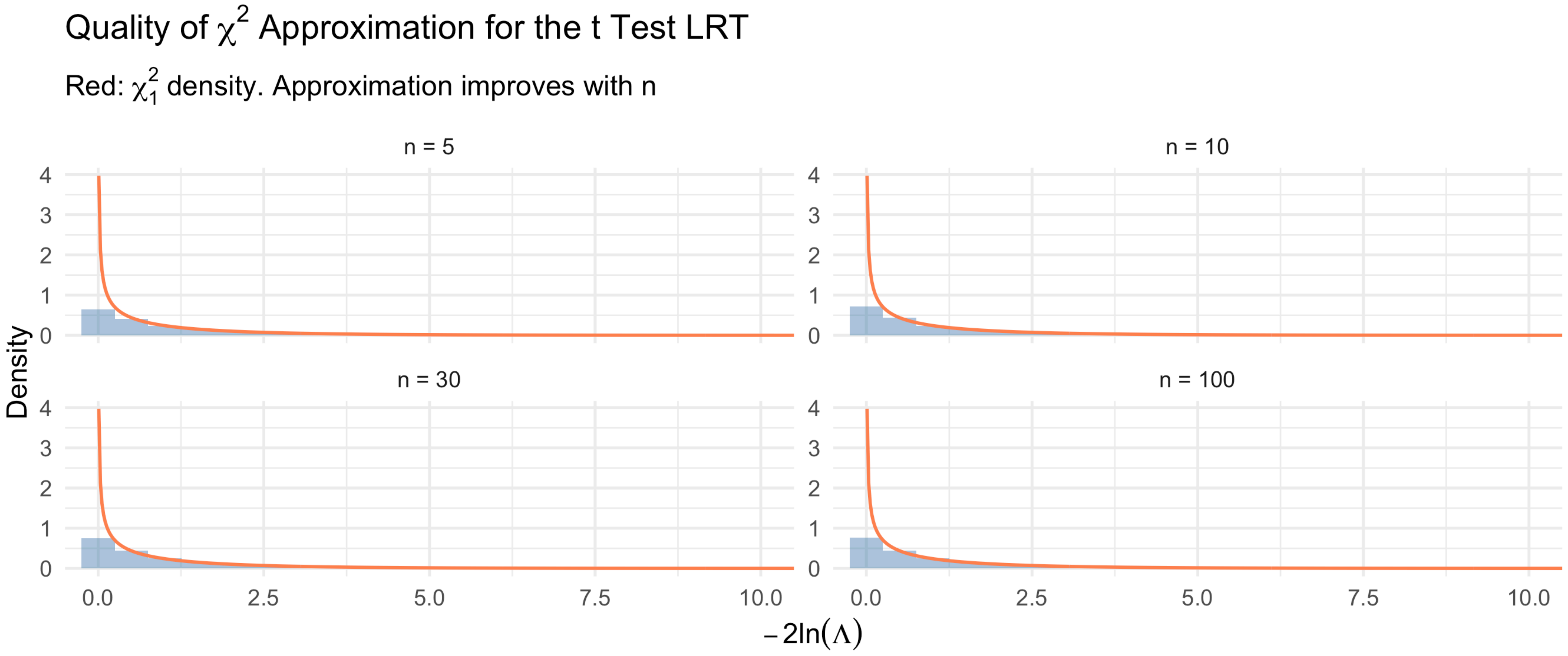
Simulation: Verifying the Chi-Squared Approximation

▶ Code



How Good is the Approximation for Small n?

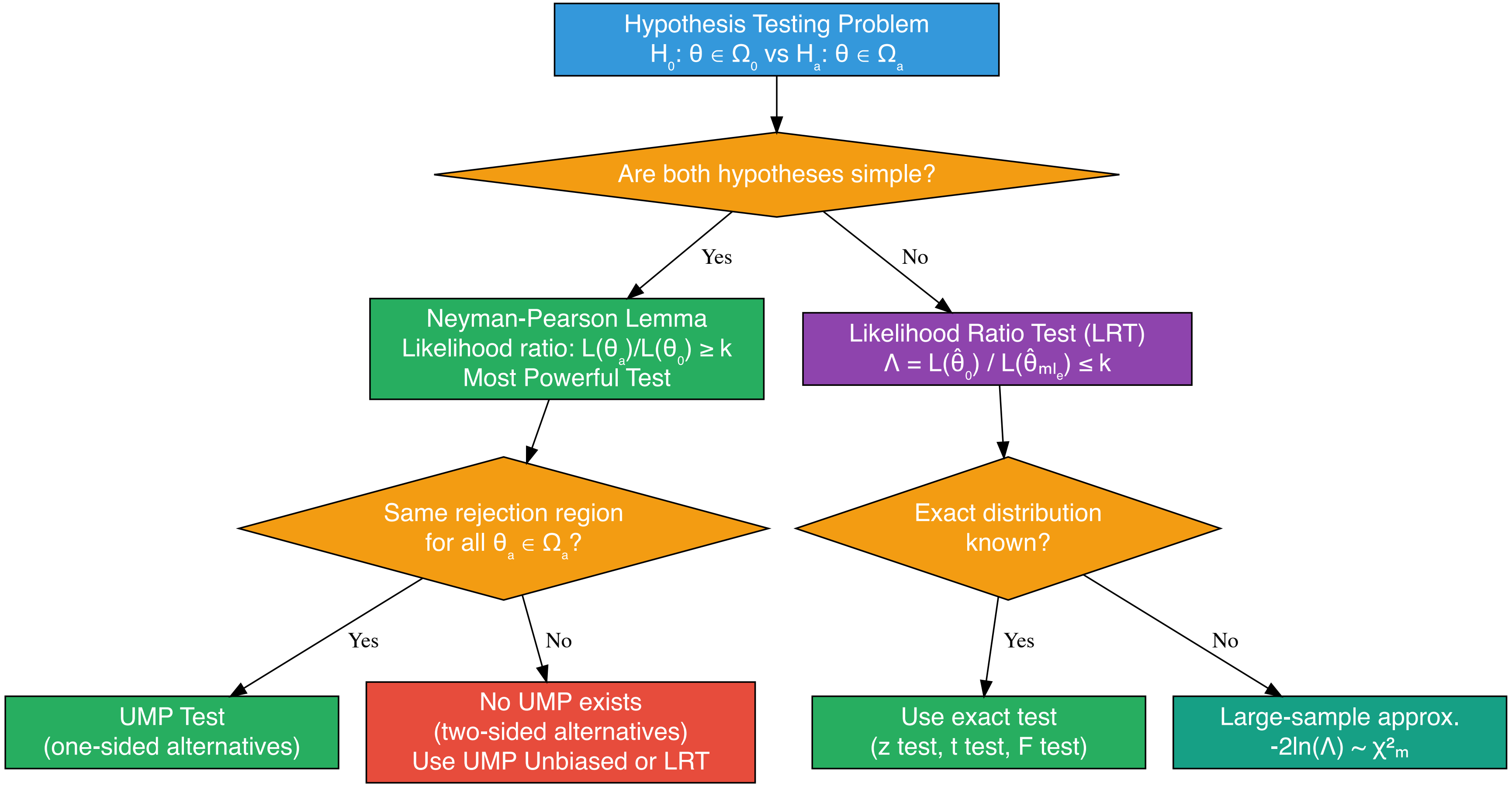
▶ Code



Connecting the Pieces

The Big Picture: Hypothesis Testing Framework

▶ Code



Summary: Classical Tests as LRTs

Test	Hypotheses	LRT Produces	Exact or Approx?
One-sample z	$H_0 : \mu = \mu_0, \sigma$ known	$z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$	Exact
One-sample t	$H_0 : \mu = \mu_0, \sigma$ unknown	$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$	Exact
Pooled two-sample t	$H_0 : \mu_1 = \mu_2$, equal σ	Pooled t statistic	Exact
Chi-squared goodness-of-fit	$H_0 : p_i = p_{i,0}$	$\sum(O_i - E_i)^2/E_i$	Large-sample
F test (ANOVA)	$H_0 : \mu_1 = \cdots = \mu_k$	F ratio	Exact (normal)

Summary: Classical Tests as LRTs

Test	Hypotheses	LRT Produces	Exact or Approx?
One-sample z	$H_0 : \mu = \mu_0, \sigma$ known	$z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$	Exact
One-sample t	$H_0 : \mu = \mu_0, \sigma$ unknown	$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$	Exact
Pooled two-sample t	$H_0 : \mu_1 = \mu_2$, equal σ	Pooled t statistic	Exact
Chi-squared goodness-of-fit	$H_0 : p_i = p_{i,0}$	$\sum(O_i - E_i)^2/E_i$	Large-sample
F test (ANOVA)	$H_0 : \mu_1 = \cdots = \mu_k$	F ratio	Exact (normal)

Key Insight

The LRT principle is a **unifying framework**. Many of the "named" tests from introductory statistics — z tests, t tests, F tests, chi-squared tests — can all be derived as likelihood ratio tests. The LRT gives us a principled way to construct new tests for situations not covered by standard formulas.

Comparison of Approaches: When to Use What

Neyman-Pearson Lemma:

- Applies to **simple vs simple** hypotheses
- Guarantees **optimality** (most powerful test)
- Useful for proving that known tests are optimal

Comparison of Approaches: When to Use What

Neyman-Pearson Lemma:

- Applies to **simple vs simple** hypotheses
- Guarantees **optimality** (most powerful test)
- Useful for proving that known tests are optimal

UMP Tests:

- Extend NP to **one-sided composite** alternatives
- Exist mainly for one-parameter exponential families with one-sided alternatives
- Do **not** exist for two-sided alternatives in general

Comparison of Approaches: When to Use What

Neyman-Pearson Lemma:

- Applies to **simple vs simple** hypotheses
- Guarantees **optimality** (most powerful test)
- Useful for proving that known tests are optimal

UMP Tests:

- Extend NP to **one-sided composite** alternatives
- Exist mainly for one-parameter exponential families with one-sided alternatives
- Do **not** exist for two-sided alternatives in general

Likelihood Ratio Tests:

- Apply to **any** testing problem (simple or composite, one or multiple parameters)
- **General-purpose recipe** that always yields a reasonable test
- Often coincide with optimal tests when those exist
- Large-sample χ^2 approximation makes them practical even when exact distributions are unknown

Key Takeaways and Looking Ahead

Key Takeaways

Today's Main Points

1. **Simple hypotheses** completely specify the distribution; composite hypotheses do not
2. **Neyman-Pearson Lemma:** the likelihood ratio test $L(\theta_a)/L(\theta_0) \geq k$ is the most powerful test for simple H_0 vs simple H_a
3. **Power function** $\pi(\theta') = P_{\theta'}(\text{Reject } H_0)$ characterizes test performance across all parameter values
4. **UMP tests** maximize power for *every* alternative; they exist for one-sided tests in exponential families
5. **Likelihood ratio test** $\Lambda = L(\hat{\theta}_0)/L(\hat{\theta}_{mle})$ provides a general recipe for composite hypotheses
6. **LRT $\Rightarrow$ classical tests:** the z test and t test are both LRTs
7. **Wilks' theorem:** $-2\ln(\Lambda) \xrightarrow{d} \chi^2_m$ provides a large-sample approximation

Key Takeaways

Today's Main Points

1. **Simple hypotheses** completely specify the distribution; composite hypotheses do not
2. **Neyman-Pearson Lemma:** the likelihood ratio test $L(\theta_a)/L(\theta_0) \geq k$ is the most powerful test for simple H_0 vs simple H_a
3. **Power function** $\pi(\theta') = P_{\theta'}(\text{Reject } H_0)$ characterizes test performance across all parameter values
4. **UMP tests** maximize power for *every* alternative; they exist for one-sided tests in exponential families
5. **Likelihood ratio test** $\Lambda = L(\hat{\theta}_0)/L(\hat{\theta}_{mle})$ provides a general recipe for composite hypotheses
6. **LRT $\Rightarrow$ classical tests:** the z test and t test are both LRTs
7. **Wilks' theorem:** $-2\ln(\Lambda) \xrightarrow{d} \chi^2_m$ provides a large-sample approximation

Common mistakes to avoid:

- Confusing the NP ratio $L(\theta_a)/L(\theta_0)$ (reject when large) with the LRT ratio $L(\hat{\theta}_0)/L(\hat{\theta}_{mle})$ (reject when small)
- Assuming UMP tests always exist (they don't for two-sided alternatives)
- Using the χ^2 approximation when n is too small
- Forgetting that the LRT degrees of freedom m count *free* parameters, not total parameters

Looking Ahead

Next class (Lesson 16): Further Aspects of Hypothesis Testing; Bootstrap Hypothesis Testing (Devore 9.6, C&H Chapter 8)

- Statistical vs practical significance
- Relationship between confidence intervals and hypothesis tests
- Bootstrap hypothesis testing
- Permutation tests

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Section 9.5
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Section 8.4
- Neyman, J. & Pearson, E. S. (1933). On the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society A*, 231, 289–337.
- Wilks, S. S. (1938). The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9(1), 60–62.

Questions?

Thank you!